21 The nth term of a sequence is $n^2 + 5$

(a) (i) Find the first two terms of this sequence.

$1^2 + 5 = 6, \quad 2^2 + 5 = 9$

1 mark
Substitute n = 1 and 2.

(1)

(ii) Is 126 a term of this sequence?
You must show how you get your answer.

$126 - 5 = 121 = 11^2$

1 mark
Check a positive integer n.

Yes: $11^2 + 5 = 126$, so it is term 11.

(1)

Here are the first five terms of an arithmetic sequence.

26 19 12 5 –2

(b) Find an expression, in terms of n, for the nth term of this sequence.

Difference = −7; $-7 \times 1 + 33 = 26$

1 mark
Use −7n.

nth term = $-7n + 33$

1 mark
Adjust to first term 26.

(2)

(Total for Question 21 is 4 marks)